

probability & statistics

* statistics *

اللهم صلي وسلم وبارك على سيدنا محمد

08

sample space: consider the experiment, A wheel with 7 numbers on perimeter spins and comes to rest so that the pointer so that the pointer points within a number sector.

→ **symbol:** S

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Ex: {1, 3, 5, 7, 6}

01

sample size (S) = 5

02

Ex: if X is a variable and $X_i = 1, 2, 3, 4$
 $X_1 = 1.2$ and $X_2 = 2.3$ and $X_3 = 4.2$ and $X_4 = 3.1$.

find when:

$$\textcircled{1} \sum_{i=1}^{n=4} X_i \quad \textcircled{2} \left[\sum_{i=1}^{n=4} X_i \right]^2$$

$$\textcircled{3} \sum_{i=1}^{n=4} (X_i - 2) \quad \textcircled{4} \sum_{i=1}^{n=4} 2 \cdot X_i$$

$$\textcircled{5} \sum_{i=1}^{n=4} \frac{1}{2} X_i - 2 \quad \textcircled{6} \sum_{i=1}^{n=4} (-2X_i + 3)$$

$$\textcircled{7} \sum_{i=1}^{n=4} \frac{X_i}{5}$$

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$$\textcircled{1} (X_1)^2 + (X_2)^2 + (X_3)^2 + (X_4)^2$$

$$= (1.2)^2 + (2.3)^2 + (4.2)^2 + (3.1)^2 = 33.98$$

$$\textcircled{2} X_1 + X_2 + X_3 + X_4 = 10.8$$

$$\therefore (10.8)^2 = 116.64$$

$$\textcircled{3} X_i - 2$$

$$\begin{array}{rcl} X_1 - 2 & + & 1.2 - 2 \\ X_2 - 2 & + & 2.3 - 2 \\ X_3 - 2 & + & 4.2 - 2 \\ X_4 - 2 & + & 3.1 - 2 \\ \hline & & 2.8 \end{array}$$

$$\textcircled{4} \sum_{i=1}^{n=4} 2 \cdot X_i = 2 \sum_{i=1}^{n=4} X_i$$

$$\textcircled{5} \sum_{i=1}^{n=4} \frac{1}{2} X_i - 2 = \frac{1}{2} \sum_{i=1}^{n=4} X_i - 2$$

$$\therefore \frac{1}{2} \times 2.8 = 1.4$$

$$\textcircled{6} \begin{array}{rcl} -2 \cdot X_1 + 3 & & -2 \cdot 1.2 + 3 \\ -2 \cdot X_2 + 3 & & -2 \cdot 2.3 + 3 \\ -2 \cdot X_3 + 3 & \Rightarrow & -2 \cdot 4.2 + 3 \\ -2 \cdot X_4 + 3 & & -2 \cdot 3.1 + 3 \\ \hline & & -17.4 \end{array}$$

$$\textcircled{7} \frac{X_1}{5} + \dots + \frac{X_4}{5} = \frac{1.2}{5} + \dots + \frac{3.1}{5} = 2.16$$

التوسطات

① * Arithmetic Mean ($\bar{x}$) المتوسط الحسابي

09 * Geometric Mean (GM) المتوسط الهندسي

10 * Harmonic Mean (HM) المتوسط التوافقي

② * Median الوسيط * Mode المنوال

* Arithmetic Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

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Ex: Find $\bar{x}$ for the data set {2, 7, 1, 3, 5}

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{2+7+1+3+5}{5} = 3.6$$

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* $\bar{x} = \frac{\sum_{i=1}^n f_i \cdot x_i}{\sum_{i=1}^n f_i}$

$x_i \rightarrow$ القيمة رقم i

$f_i \rightarrow$ مركز القيمة رقم i

Ex: calculate the mean for the following data

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	5	2	3	8	2

Marks	f_i	x_i	$f_i \cdot x_i$
0-10	5	5	25
10-20	2	15	30
20-30	3	25	75
30-40	8	35	280
40-50	2	45	90
SUM	20	125	500

$$x_i = \frac{0+10}{2} = 5 \quad x_i = \frac{10+20}{2} = 15$$

$$\bar{x} = \frac{\sum_{i=1}^n f_i \cdot x_i}{\sum_{i=1}^n f_i} = \frac{500}{20} = 25$$

Ex: if the mean of the data

$\{1, 2, 3, x, 3x-1\}$ is the 5 $x = ?$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\frac{1+2+3+x+3x-1}{5} = 5 \quad \therefore 5 + 4x = 25$$

$$\therefore 4x = 20 \quad \therefore x = 5$$

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* Geometric mean (GM)

$$GM = \sqrt[n]{\prod_{i=1}^n x_i} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

Ex: Find the mean of this data

Year	2000	2004	2008	2012	2016
Total Votes	50.3	55.7	57.1	54.9	60.1

$$GM = \sqrt[5]{50.3 \times 55.7 \times 57.1 \times 54.9 \times 60.1}$$

$$= 55.5257$$

* HARMONIC MEAN

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

EX: Find HM for the data {3, 7, 2}

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}} = \frac{3}{\frac{1}{3} + \frac{1}{7} + \frac{1}{2}} = 3.07$$

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EX: compare between the efficiency of the two machines for the 3 procedures in the following table:

	procedure	procedure	procedure
machine A	7	80	2100
machine B	3	94	2350

* for comparing between efficiency, we use the GM

$$GM = \sqrt[n]{\prod_{i=1}^n x_i}$$

$$GM_A = \sqrt[3]{7 \times 80 \times 2100} = 105.55$$

$$GM_B = \sqrt[3]{3 \times 94 \times 2350} = 87.18$$

machine A is more efficient than machine B

* median

① ترتيب البيانات (من الأصغر إلى الأكبر)

② إيجاد القيمة الوسطى (التي في المنتصف)

* odd عدد الأرقام فردى

$$\text{median} = \frac{n+1}{2}$$

* even عدد الأرقام زوجى

$$MD = \frac{n}{2} \text{ و } \frac{n}{2} + 1$$

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Ex: The data represent the number of compressive

strengths for samples of concrete are

713, 300, 618, 595, 311, 401, and 292 find

the median.

292, 300, 311, 401, 595, 618, 713

$$MD = \frac{7+1}{2} = 4 \therefore \text{Hence, the MD 401 Room}$$

The data represent the number of compressive

strengths for samples concrete are 1, 7, 3, 2, 4,

3
1, 2, 3, 3, 4, 7

$$MD = \frac{6}{2}, \frac{6+1}{2}$$

Hence the MD 3, 4 Rooms

البيانات المبوبة
Median

$$MD = L + \left[\frac{\frac{N}{2} - C}{F} \right] \cdot h$$

L = Lower Limit الفئة الوسطية

N = total number of the population of sum
مجموع التكرارات

C = cumulative frequency
مجموع التكرارات قبل الفئة الوسطية

h = length of each group طول الفئة

f تكرار الفئة الوسطية نفسها

N = مجموع التكرارات

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categories

frequency

C

20-10	الفئة	4	4
30-20	الوسطية	9	13
40-30		15	28
50-40	L	8	36
60-50		4	40

sum

$$\sum F = 40$$

$$h = 10 - 20 = 10 \quad \therefore h = 10$$

$$\frac{N}{2} = \frac{40}{2} = 20$$

$$L = 30$$

$$C = 13$$

$$F = 15$$

$$MD = 30 + \left(\frac{20 - 13}{15} \right) \times 10$$

* mode

الموالات

(الكثير رقم التكرار)

EX: {2, 2, 3, 3, 4, 5, 3, 2}

2, 3

EX: {red, blue, blue, black}

blue

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* mode

البيانات المتصلة

$$\text{mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \cdot h$$

f_1 PREVIOUS FREQUENCY VALUE FOR f_1
القيمة السابقة لتكرار f_1

f_0 MOST FREQUENT VALUE FOR f_1
التكرار السابق لـ f_1

f_2 NEXT FREQUENCY VALUE FOR f_1
التكرار التالي لـ f_1

L VALUE FOR f_1
الحد الأدنى للقياس المتوالي

find the mode of the data

marks

No. of students

10-20-30-40-50

marks

students

10-20	5	f_0	5
20-30	12	f_1	12
30-40	8	f_2	8
40-50	5	h	10
Mode = $20 + \left[\frac{12-5}{2 \times 12 - 5 - 8} \right] \times 10$		L	20

= 26.36 marks

Ex: Complete if the data has $L=20, f_1=10,$

$f_0=7, h=5$ And the mode = 27 then

f_2 will equal?

$$\text{mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \cdot h$$

$$27 = 20 + \left[\frac{10 - 7}{2 \times 10 - 7 - f_2} \right] \times 5$$

$$27 = 20 + \left[\frac{3}{13 - f_2} \right] \times 5$$

$$7 = \left[\frac{3}{13 - f_2} \right] \times 5$$

$$\frac{7}{5} = \frac{3}{13 - f_2}$$

$$7(13 - f_2) = 15$$

$$91 - 7 \cdot f_2 = 15$$

$$-7 \cdot f_2 = -79$$

$$\therefore f_2 = 10.857$$

measure of dispersion (quartile)

→ Q_1 :

for odd data $Q_1 = \frac{n+1}{4}$

for even data $Q_1 = \frac{n}{4}, \frac{n}{4} + 1$

for TEBULER $Q_1 = L + \left[\frac{\frac{n}{4} - C}{f} \right] \cdot h$

$IR = Q_3 - Q_1$

$SIR = \frac{1}{2} IR$

Let the data set be $\{4, 7, 1, 5, 3\}$ Find

$Q_1 = \frac{5+1}{4} = 1.5$

$Q_3 = \frac{3 \times 6}{4} = 4.5$

$IR = Q_3 - Q_1$

$4.5 - 1.5 = 3$

$SIR = 3 \times \frac{1}{2} = 1.5$

→ Q_2 $Q_2 = \frac{n}{2}$

even $= \frac{n}{2}, \frac{n}{2} + 1$

$Q_2 = L + \left[\frac{\frac{n}{2} - C}{f} \right] \cdot h$

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Let the data set be $\{2, 3, 4, 7, 11\}$ And

Q_1, Q_2, Q_3

$Q_1 = \frac{n}{4}, \frac{n}{4} + 1 = \frac{5}{4}, \frac{7}{4} = [1.5, 2.5]$

$[2, 3] \quad Q_1 = \frac{2+3}{2} = 2.5$

$Q_2 = \frac{n}{2}, \frac{n}{2} + 1 = \frac{5}{2}, \frac{6}{2} + 1 = 3, 4$

$Q_2 = \frac{3+4}{2} = 2.5$

$Q_3 = \frac{3n}{4}, \frac{3(n+1)}{4} = \left[\frac{3 \times 5}{4}, \frac{3 \times 6}{4} \right]$

$= [4.5, 5.5] \sim [5, 6] = \frac{5+6}{2} = 5.5$

→ Q_3 :

$$\text{odd } Q_3 = \frac{3(n+1)}{4}$$

$$\text{even } Q_3 = \frac{3n}{4}, \frac{3n}{4} + 1$$

$$Q_3 = L + \left[\frac{\frac{3n}{4} - c}{F} \right] h$$

EX:

class f c

01	0 -	7	7
	10 -	9	16
02	20 -	12	28
	30 -	21	49
03	40 -	39	88
	50 -	44	132
04	60 -	29	161
	70 -	18	179
05	80 -	14	193
	90 - 100	7	200

$$Q_1 = \frac{200}{4} = 50$$

$$Q_2 = \frac{200}{2} = 100$$

$$Q_3 = \frac{3 \times 200}{4} = 150$$

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class

f

c

0-10

7

7

10-20

9

16

20-30

12

28

L1 30-40

21

(49) - C1

(40-50)

(39) - F1

(88) - C2

L2 50-60

(44) - F3

(132) - C3

L3 60-70

(289) - F3

161

70-80

18

179

80-90

14

193

90-100

7

200

h=10

Q1 Q2 Q3

L 40 50 60

F 39 44 29

C 49 88 132

h 10 10 10

measures of variability

08 → The variance and standard deviation for scattered data
09 → ungrouped data بيانات غير مبنوه

10 → The variance (σ^2)

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \text{ or } \frac{\sum_{i=1}^n x_i^2 - \bar{x}^2}{n}$$

01 → The standard deviation (σ)

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} \text{ or } \sqrt{\frac{\sum_{i=1}^n x_i^2 - \bar{x}^2}{n}}$$

Ex:

find the variance and standard deviation of the set of data 66, 73, 69, 74, 68

--- solution ---

10 Variance (σ^2) = $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} = \frac{\sum_{i=1}^n x_i^2 - \bar{x}^2}{n}$

12 $\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{66+73+69+74+68}{5} = 70$

01 $\sigma^2 = \frac{1}{5} (66^2 + 73^2 + 69^2 + 74^2 + 68^2) - 70^2$

03 $\sigma^2 = 9.2$ $\therefore \sigma = \sqrt{9.2} = 3.03$

coefficient of variation (C.V) معامل التغير

$$C.V = \frac{\sigma}{\mu} \times 100$$

μ → المتوسط الحسابي

σ → الانحراف المعياري

Ex: You have $\mu = 168,6$ and $\sigma^2 = 4.3766$ of a set of data then the coefficient of variation equals

$$CV = \frac{\sigma}{\mu} \times 100 \quad \therefore \sigma = \sqrt{4.3766} = 2.092$$

$$= \frac{2.092}{168,6} \times 100 = 1,2408$$

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* INTERMEDIATE STATISTICS

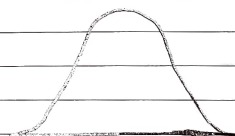
SKENNESS



POSITIVE SKEW

$$SK > 0$$

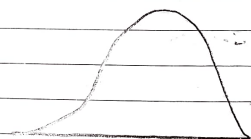
$$SK > \bar{X}$$



SYMMETRICAL DISTRIBUTION

$$SK = 0$$

$$SK = \bar{X}$$



NEGATIVE SKEW

$$SK < 0$$

$$SK < \bar{X}$$

$$SK = \frac{3(\bar{X} - \text{Median})}{\sigma}$$

$$SK = \frac{\bar{X} - \text{mode}}{\sigma}$$

Ex: calculate the skewness coefficient for the data set: {66, 73, 69, 74, 68}

$$\bar{x} = \frac{66 + 73 + 69 + 74 + 68}{5} = 70$$

$$\text{median} = \{66, 68, \underline{69}, 73, 74\} \Rightarrow \text{odd}$$

$$\text{median} = 69$$

$$\sigma^2 = \frac{1}{5} [24546 - 70^2] = 9.2$$

$$\sigma = \sqrt{9.2} = 3.03$$

$$SK = \frac{3(\bar{x} - \text{median})}{\sigma} = \frac{3(70 - 69)}{3.03} = 0.99$$

calculate the skewness coefficient for the data set: {1, 2, 3, 4, 1}

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 1}{5} = 2$$

$$\{1, 1, \underline{2}, 3, 4\}$$

↓
odd

$$\text{median} = 2$$

$$\text{mode} = 1$$

$$\sigma^2 = \frac{1}{4} [31 - 2^2] = 1.5$$

$$\sigma = \sqrt{1.5} = 1.225$$

$$SK = \frac{\bar{x} - \text{mode}}{\sigma} = \frac{2 - 1}{1.225} = 0.816$$

* mean deviation

$$MD = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

Ex: Find the MD of the data

{6, 5, 7, 7, 8, 9, 9, 5}

$$\bar{x} = 7$$

$$MD = \frac{1}{8} [|6-7| + |5-7| + |7-7| + |8-7| + |9-7| + |9-7| + |5-7|]$$

$$= 1.25$$

moment about zero

$$m_1(o) = \frac{\sum x_i}{n}$$

$$\sum_3(o) = \frac{\sum x^3}{n}$$

$$m_2(o) = \frac{\sum x^2}{n}$$

$$\sum_4(o) = \frac{\sum x^4}{n}$$

* moment about mean

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$$m_1 = 0$$

$$m_2 = m_2(o) - [m_1(o)]^2 = 0^2$$

$$m_3 = m_3(o) - 3m_1(o)m_2(o) + 2[m_1(o)]^3$$

$$m_4 = m_4(o) - 4m_1(o)m_3(o) + 6[m_1(o)]^2m_2(o) - 3[m_1(o)]^4$$

* kurtosis coefficient

* skewness coefficient

$$k_u = \frac{m_4}{\sigma^4}$$

$$s_k = \frac{m_3}{\sigma^3}$$

Ex: Find the 1st, 2nd, 3rd, 4th moments for

the data set 1, 2, 3, 3, 4, 4, 4, 4, 5, 5, 5, 6, 7

and then calculate the coefficient of

kurtosis.

x_i	x_i^2	x_i^3	x_i^4
1	1	1	1
2	4	8	16
3	9	27	81
3			
4			
4			
4			
4			
5			
5			
5			
6			
7	49		2401
Σ	60	272	1344
			7112

$$m_1(0) = \frac{\sum x_i}{n} = \frac{60}{15} = 4$$

$$m_2(0) = \frac{\sum x_i^2}{n} = \frac{272}{15} = 18.13$$

$$m_3(0) = \frac{\sum x_i^3}{n} = \frac{1344}{15} = 89.6$$

$$m_4(0) = \frac{\sum x_i^4}{n} = \frac{7112}{15} = 474.13$$

$$KV = \frac{m_4}{\sigma^4}$$

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$$\sigma^2 = \sqrt{m_2(0) - [m_1(0)]^2} = \sqrt{18.13 - 16} = 1.46$$

$$m_4 = m_4(0) - 4m_1(0)m_3(0) + 6[m_1(0)]^2m_2(0) - 3[m_1(0)]^4$$

$$= 13.01$$

$$KV = \frac{m_4}{\sigma^4}$$

Skewness coefficient

$$SK = \frac{m_3}{\sigma^3}$$

$$m_3 = m_3(0) - 3m_1(0)m_2(0) + 2[m_1(0)]^3$$

$$= 7$$

$$\therefore SK = \frac{7}{1.46^3} = 2.25$$

$$r_{xy} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sigma_x \sigma_y} \Rightarrow \text{معامل الارتباط}$$

$$= \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}}$$

r_{xy}	Correlation degree
0	No correlation لا يوجد ارتباط
$0 < r_{xy} < 0.5$	Weak الارتباط ضعيف
$r_{xy} = 0.5$	Moderate متوسط
$0.5 < r_{xy} < 1$	Strong قوي
$r_{xy} = 1$	Perfect تام

→ TYPE OF CORRELATION

$r_{xy} \rightarrow (+)$ Then the correlation between X and Y is directly correlation الارتباط طردي

$r_{xy} \rightarrow (-)$ Then the correlation between X and Y is ~~directly~~ inversely correlation الارتباط عكسي

* COVARIANCE التباين المشترك

$$c_{xy} = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

* SPEARMAN RANK CORRELATION معامل ارتباط بيرسون للترتيب

$$r = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

Ex: The following table presents two sets:

x_i	25	26	24	23	27	21	24	27	22	24
y_i	49	49	46	46	47	43	47	48	43	46

calculate the correlation coefficient and covariance.

Solution

$n = 10$

	x_i	y_i	x_i^2	y_i^2	$x_i y_i$	$(x_i - \bar{x})$	$(y_i - \bar{y})$
02	25	49					
	26	49					
02	24	46					
	23	46					
04	27	47					
	21	43					
05	24	47					
	27	48					
06	22	43					
	24	46					
Σ	243	463					

$n = 10$

$$r_{xy} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n \sum x_i^2 - (\sum x_i)^2} \sqrt{n \sum y_i^2 - (\sum y_i)^2}}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{y} = \frac{\sum y_i}{n}$$

$$C_{xy} = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

2: Spearman Rank Correlation :

X	Y	Rank X	Rank Y	d	d ²
25	49	7	9.5	-2.5	6.25
26	49	8	9.5	-1.5	2.25
24	46	5	4	1	1
23	46	3	4	-1	1
27	47	9.5	6.5	-3	9
21	43	1	1.5	-0.5	0.25
24	47	5	6.5	-1.5	2.25
27	48	9.5	8	1.5	2.25
28	43	2	1.5	0.5	0.25
24	46	5	4	1	1

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Note :

يتم ترتيب قيم X أو Y من الأصغر إلى الأكبر

- 1. يتم ترتيب قيمة تأخر الرتبة
- 2. القيمة القاطعة تأخذ

التعامل مع القيم المتكررة (التعادل)

إذا تكررت نصف القيمة المتكررة من بين إعطائها متوسط الرتب المتوائمة ستشغلها

مثال : القيمة (24) تكررت 3 مرات = 4, 5, 6

$$\therefore \frac{4+5+6}{3} = 5$$

وهكذا الأمر بالنسبة لـ Y.

→ ~~هنا~~ هونا ج الفرق بين رتبة Y ورتبة X

$$(Rank X - Rank Y)$$

Probability

$\{1, 2, 3, 4, 5, 6\}$

Ex: Shown throws die 400 times

and record the score of getting 5's

30 times what could be the

probability is:

① getting score of 5?

② getting a score under 5?

① $\frac{30}{400} = \frac{3}{40}$

② $1 - \frac{3}{40} = \frac{37}{40} = \frac{37}{40}$

#

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$* P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A^c \cap B^c) = P(A \cup B)^c = 1 - P(A \cup B)$$

$$P(A^c \cup B^c) = P(A \cap B)^c = 1 - P(A \cap B)$$

Note:

$$\emptyset^c = S$$

$$A \cup A^c = S$$

$$A \cap A^c = \emptyset$$

$$S^c = \emptyset$$

اللهم صلي وسلم وبارك على سيدنا محمد